

Cheap Talk with Costly Verification

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Abstract

A sender, who is privately informed about the state of the world, sends a message to a receiver. After observing the message, the receiver decides whether to pay a cost, the size of which is private information, to learn whether the sender's message is true. I show that any influential equilibrium of the model is characterized by a threshold: the sender's expected utility from lying. If the state is above the threshold, the sender tells the truth, and if the state is below the threshold, the sender lies by mixing over the messages above the threshold. In response, the receiver verifies all messages above the threshold with a positive probability. I then show the receiver's expected utility is higher due to verification as a result of two effects: an informational effect and a deterrence effect. Moreover, the deterrence effect can increase the expected informational effect of verification.

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1 Introduction

Consider the problem of a politician meeting with a lobbyist who advocates for an energy policy by citing evidence that similar policies have succeeded in the past. Recognizing the lobbyist’s biased preferences—the lobbyist wants the politician to support the policy regardless of whether it will succeed—the politician might wish to verify the lobbyist’s claims by directing a staffer to investigate further. However, verification is costly: it requires time and attention that could be devoted to the staffer’s other responsibilities. Thus, the politician may choose not to undertake verification either because the cost is high (e.g., the office is very busy) or the informational value of learning whether the lobbyist was honest is small.

Now consider the problem from the lobbyist’s perspective: they want to convince the politician to implement the policy, but the politician only wants to implement it if it will succeed. One approach the lobbyist might take is to claim there is evidence that shows similar policies have succeeded elsewhere. But what if the record of past implementations is actually mixed or even bad? In choosing whether to lie to the politician, the lobbyist must consider two possibilities—what the politician will believe if the claim is verified and what the politician will believe if it is not—as well as the probability of verification. In turn, the probability of verification may depend on the content of the lobbyist’s message as well as the politician’s cost of verification—something that the lobbyist does not observe.

This paper studies communication between a biased sender and a receiver who can verify the sender’s message by paying a privately known cost. While the lobbyist-politician example is a natural application, the framework extends to many other strategic interactions in politics. A candidate may try to persuade voters of her qualifications, for instance, and voters can independently verify her claims; likewise, an incumbent politician may tout the success of a recently implemented policy, and voters may turn to journalists or fact-checkers to assess whether the claimed success is real. Across these examples, I focus on three unifying features. First, the receiver can verify the truthfulness of the sender’s message (e.g., the politician can verify whether the lobbyist was honest when saying similar policies had succeeded). Second,

verification speaks to the truthfulness of the sender’s message rather than the underlying state of the world (e.g., the politician can learn the lobbyist lied but not whether the policy will succeed if implemented). Third, the sender does not know how costly it will be for the receiver to verify the message (e.g., the lobbyist does not know how busy the politician’s office is).

At the beginning of the game studied in this paper, a sender (she) privately learns the state of the world, which is drawn from a finite subset of $[0, 1]$. Then, she communicates with a receiver (he) using cheap talk. After observing the sender’s message, the receiver chooses whether to verify it. If he does, he pays a privately known cost and learns whether the message is true, but does not learn the state of the world. If he does not, he learns nothing about the veracity of the sender’s message. Then, the receiver chooses an action that is payoff relevant to both players.

I begin by showing that any influential equilibrium of the game has a straightforward structure characterized by a threshold: the sender’s expected utility in equilibrium when she lies. When the state is above this threshold, the sender tells the truth, and when the state is below this threshold, she lies by randomizing over the messages that are above the threshold. In turn, the receiver verifies all messages above this threshold with a positive probability.

This threshold structure points to a dependence between endogenous verification and suspicion of lying that extends to all messages in the support of the sender’s strategy (except when the expected utility from lying is equal to one of the states). If the receiver believes a particular message in the support of the sender’s strategy is only sent truthfully, he has no incentive to verify it. But, then there must be a deviation: either the expected utility of lying is below that message—in which case there is a profitable deviation from lying to sending that message—or the expected utility of lying is above that message—in which case there is a profitable deviation from telling the truth with that message to lying. Hence, the usefulness of endogenous verification as a way to support informative communication requires suspicion about the truthfulness of the messages conveyed by the sender. Moreover,

this is true for all messages in the support of the sender’s strategy—endogenous verification does not produce entirely honest messages.

That all messages in the support of the sender’s strategy are sent truthfully and as lies is a key difference between the environment studied here and existing cheap talk models with verification. In models with endogenous verification and vague messages (e.g., [Sadakane and Tam, 2026](#)), as well as models with stochastic verification (e.g., [Dziuda and Salas, 2018](#)), some intermediate messages are only sent truthfully. In the former case, this is because with sufficiently punishing off the equilibrium path beliefs, a lying sender can be made indifferent between lying and sending a vague but truthful message. In the latter case, verification does not depend on the receiver’s endogenous choice, so even messages that are only sent truthfully are verified.

Returning to the lobbyist example, these models predict that there are some intermediate messages the lobbyist will only send truthfully—for instance, claiming past implementations were “somewhat successful.” In contrast, in my model, there are no “safe” messages that are only sent truthfully. If the lobbyist only says past implementations were “somewhat successful” when it is true, a lobbyist who advocates for a policy by lying about its absence of past success will have a deviation from her equilibrium strategy to saying the policy was “somewhat successful,” as this deviation must yield a better payoff than lying according to her equilibrium strategy.

I then solve for all influential equilibria of the model where there are three equally likely states and the cost of verification is uniform. Two distinct types of equilibria exist. In the first, the highest-type sender tells the truth, while the two lower-type senders lie by pretending to be the highest type. In the second, the two highest-type senders tell the truth, and the lowest-type sender lies by randomizing between lying with the highest types’ equilibrium messages.

Analysis of the three-state case illustrates that there are two effects of verification. The first is the direct *informational* effect, which is the difference between the variance of the

receiver’s belief when he does not verify the sender’s message and the expected variance of his belief when he does. The second is an indirect *deterrence* effect of verification: verification can deter the intermediate type from lying due to an endogenous cost that emerges in equilibrium, the fear of being pooled with the lowest type.

While verification is costly, the receiver only chooses to verify when the informational effect of verification exceeds his cost. Thus, the receiver strictly benefits from the informational effect of verification, which is present in all influential equilibria. Moreover, he benefits from the deterrence effect, which is present in some influential equilibria—in particular, the influential equilibria where the intermediate type tells the truth. Thus, the improvement in the receiver’s *ex ante* expected utility from verification is always due to the informational effect and sometimes due to the additional deterrence effect. This observation contrasts with existing work where the informational effect is completely offset by the cost of verification, which is known, and thus the increased *ex ante* expected utility enjoyed by the receiver due to verification comes entirely from the deterrence effect ([Sadakane and Tam, 2026](#)).

One might conjecture that the informational and deterrence effects act like substitutes: the greater the deterrence effect, the smaller the informational effect. However, I conclude by showing that this intuition is not always correct. The key to understanding this is that when verification reveals the truthfulness of the sender’s message but not the state—as is assumed—the deterrence effect shapes the effectiveness of verification. We can see this in the three-state example. When the intermediate type is deterred from lying, verification perfectly reveals the state. In contrast, when the intermediate type lies, the receiver is still uncertain about the state when he verifies a message and learns it is a lie. So deterrence can improve verification. Thus, under certain circumstances—the intermediate state is sufficiently far from the lowest state and the upper bound of the distribution of the cost to verify is sufficiently large—the deterrence effect increases the informational effect of verification.

2 Related Literature

This paper studies a model of cheap talk with endogenous verification. In the seminal paper on cheap talk, [Crawford and Sobel \(1982\)](#) show that when the sender is sufficiently biased, no information transmission occurs in equilibrium. This paper joins others on cheap talk with verification in showing that if the receiver can detect a lie sent by the sender—either with an exogenous probability (e.g., [Dziuda and Salas, 2018](#); [Holm, 2010](#)) or, as in this paper, through an endogenous decision by the receiver (e.g., [Sadakane and Tam, 2026](#); [Levkun, 2025](#); [Ball and Gao, 2025](#))—influential communication arises.¹

In [Sadakane and Tam \(2026\)](#), the paper closest to mine, the receiver chooses whether to incur a publicly known cost to inspect a message sent by a privately informed sender. In equilibrium, the ability of the sender to send “vague” messages and punishing off the equilibrium path beliefs mean that some messages are only sent truthfully and thus are not verified. In contrast, in my model, less punishing off the equilibrium path beliefs and a coarser message space mean that in any influential equilibrium of the model—except for a knife-edge case—all messages in the support of the sender’s strategy are sent truthfully and as lies and are verified with a positive probability. In both models, the receiver’s ability to verify the sender’s message has two effects: an informational effect, which reveals whether the sender’s message is true, and a deterrence effect, which may deter a sender of an intermediate type from lying. However, in [Sadakane and Tam \(2026\)](#), the cost of verification completely offsets the informational effect. In contrast, when the receiver has private information about his cost, the informational effect often strictly exceeds the realized cost the receiver must pay to verify the sender’s message. Thus, my model points to a more positive view of the informational value of endogenous verification. Moreover, I show that in some cases the deterrence effect can increase the expected informational effect.

My model is also related to [Dziuda and Salas \(2018\)](#), in which a biased sender commu-

¹Less related to this model, others study exogenously detected deceit in other communication frameworks like Bayesian persuasion (e.g., [Ederer and Min, Forthcoming](#); [Venkatesh et al., 2025](#)) and detectable lying when the sender’s bias is limited (e.g., [Balbuzanov, 2019](#))

nicates with a receiver who learns whether the sender’s message is true with an exogenous probability. By contrast, in my model, the receiver endogenously chooses whether to verify the sender’s message. This choice produces a distinct equilibrium structure. In [Dziuda and Salas \(2018\)](#), some intermediate messages are only sent truthfully, whereas, in my model, all messages in the support of the sender’s strategy are sent truthfully and as lies. When verification is exogenous, messages that are only sent truthfully are still verified with a positive probability; this threat of verification prevents the sender from deviating to lying with some intermediate messages. When verification is endogenous, however, messages that are only sent truthfully on the equilibrium path are never verified, giving the sender a profitable deviation when these messages are higher than the sender’s expected utility from lying in equilibrium—which is the case in any influential equilibrium of my model. Hence, my model illustrates a dependence between verification and the expectation of lying that extends to every message sent in equilibrium.

This paper is also related to models of cheap talk with an exogenous cost of lying. This cost might arise due to a psychological cost associated with lying (e.g., [Kartik et al., 2007](#); [Minozzi and Woon, 2013](#)) or the existence of receivers who take the sender’s message at face value (e.g., [Kartik et al., 2007](#)).² In both cases, deception emerges in equilibrium, but the cost of lying constrains the sender to the extent that information can be conveyed despite disagreement between the sender and receiver(s). In my model, the sender does not pay an exogenous cost to lie. Yet, I show that in equilibrium, an endogenous cost of lying emerges. This cost is the source of the deterrence effect of verification.

Finally, in its applications, this paper is related to the literatures on fact-checking and informational lobbying. The former literature is primarily empirical (e.g., [Weeks and Garrett, 2014](#); [Weeks, 2015](#); [Nyhan and Reifler, 2010](#)), but includes some theoretical work (e.g., [Levkun, 2025](#)). Of particular relevance to my paper are [Nyhan and Reifler \(2015\)](#) and [Lim \(2018\)](#), who document empirical evidence of a deterrence effect of fact-checking on politi-

²To a lesser degree, my model is related to a broader literature of strategic communication where lying is costly (e.g., [Nguyen and Tan, 2021](#); [Guo and Shmaya, 2019](#)).

cians’ behavior, and [Gottfried et al. \(2013\)](#) and [Pingree et al. \(2014\)](#), who document empirical evidence of an informational effect of fact-checking. The latter literature encompasses theoretical (e.g., [Ellis and Groll, 2020](#)) and empirical work (e.g., [Hojnacki and Kimball, 1998](#)). My paper is closest to [Potters and Van Winden \(1992\)](#) and [Austen-Smith and Wright \(1992\)](#), which analyze settings in which a biased lobbyist is able to convey her private information to a politician who recognizes the lobbyist is biased. However, my setting does not rely on information acquisition or signals being costly.

3 Model

There is a sender (S , “she”) and a receiver (R , “he”). At the beginning of the game, the sender privately learns the state of the world, her “type”, $\theta \in \Theta$, where Θ is a finite subset of $[0, 1]$, each $\theta \in \Theta$ occurs with probability $h(\theta) > 0$, $N = |\Theta|$, and $\theta \in \Theta$ are indexed such that $\theta_1 < \dots < \theta_N$. The sender communicates with the receiver by choosing a message m from the message space $M = \Theta$. If $m = \theta$, the message is “true”—the sender tells the truth—and if $m \neq \theta$, the message is “false”—the sender lies.³

After observing the sender’s message, the receiver decides whether to verify it ($c = 1$) or not ($c = 0$). Verification is costly; if the receiver verifies the sender’s message, he pays a cost $\beta\kappa$ where $\kappa \in K = [0, 1]$ is private information he learns at the start of the game and $\beta > 0$ is a parameter observed by both players. I assume κ has a prior distribution g , where $g(\kappa) > 0$ for all $\kappa \in [0, 1]$ and G is the associated CDF. If the receiver verifies the sender’s message, he learns whether it is true ($v = t$) or false ($v = f$) but does not learn the state. If he does not verify, he learns nothing ($v = \emptyset$). Finally, the receiver selects an action $a \in [0, 1]$.

Preferences. The receiver has a utility function $u_R(\kappa, \theta, a) = -(\theta - a)^2 - c \cdot \beta\kappa$, and the sender has a utility function $u_S(a) = a$. Note, the receiver’s payoff depends on the state—in

³Since the sender is restricted to choosing a single message rather than allowed to choose a subset of the message space (e.g., $m = \{\theta_j, \theta_k\}$), these definitions are equivalent to the definitions in [Sobel \(2020\)](#).

particular, he wants to choose the action that matches the state—while the sender wants the receiver to choose $a = 1$.

Equilibrium. A (mixed) strategy for the sender is a probability function $\sigma(\cdot|\theta) : \Theta \rightarrow \Delta M$.⁴ A message m is “on the equilibrium path” or in “the support of” the sender’s strategy if $\sigma(m|\theta) > 0$ for at least one θ , and is “off the equilibrium path” otherwise. A strategy for the receiver is a tuple (c, a) where $c : K \times M \rightarrow \{0, 1\}$ and $a : M \times \{t, f, \emptyset\} \rightarrow [0, 1]$.⁵ The game is solved for perfect Bayesian equilibrium (henceforth, an “equilibrium”), where a PBE is a triple (σ, c, a) and a belief assessment such that:

1. If $m \in M$ is in the support of $\sigma(m|\theta)$, m maximizes the sender’s expected utility taking (c, a) as given.
2. If (c, a) is chosen, $c(\kappa, m)$ maximizes the receiver’s expected utility given his posterior belief over θ induced by m and $a(m, v)$ maximizes his expected utility given his posterior belief over θ induced by (m, v) .
3. Beliefs are generated using Bayes’ rule on the equilibrium path.

Throughout the paper, I focus on influential equilibria (henceforth, IEs).

Definition 1. *An equilibrium is influential if $a^*(m, v)$ is not constant on the equilibrium path (Sobel, 2013).*

The restriction to IEs only excludes babbling equilibria. In these equilibria, the sender’s messages are uninformative, and the receiver ignores them, choosing not to verify them and taking action $a^*(m, \emptyset) = \mathbb{E}[\theta]$.

⁴ $\Delta(X)$ denotes the space of lotteries over X .

⁵I restrict attention to equilibria where the receiver plays a pure strategy. This is without loss of generality as the receiver has a unique best response to any posterior belief about the sender’s type due to the strict concavity of his utility function.

Assumptions. As in many cheap talk games, the set of equilibria is large. In light of this, I make the following assumptions to refine the set of equilibria.

Assumption 1. *When lying, the sender’s strategy is independent of θ .*

In any equilibrium, when the sender lies, she must be indifferent between any lies in the support of her strategy. Moreover, since verification only reveals whether the sender’s message is true, her payoff from sending a particular message as a lie cannot depend on her type. Put differently, her expected utility from lying with message m cannot depend on whether her type is θ or θ' .

However, this fact about any equilibrium does not imply that different types of senders must use the same lying strategy. In principle, it is possible for multiple types of senders to lie and, in particular, to mix over the set of messages sent as lies using different probabilities, while remaining indifferent across the messages. Assumption 1 rules out such equilibria by requiring that the sender’s strategy when lying is independent of her type. In the Supplementary Appendix, I provide an example of an IE ruled out by Assumption 1 in which there is a message that is only sent as a lie. Thus, this assumption plays a non-trivial role in the structure of IEs studied in this paper.

Assumption 2. *If there is a message m that is not in the support of the sender’s strategy, then upon seeing m , the receiver believes $\theta = m$.*

This assumption can be justified by assuming that if a message is off the equilibrium path, there is an infinitesimally small probability the truth is accidentally revealed by the sender through a slip of the tongue (e.g., [Hart et al., 2017](#)).

The import of this assumption is that it rules out IEs supported by extremely punishing off the equilibrium path beliefs. For instance, one could assume that any sender who deviates to a message that is off the equilibrium path is the lowest type. In contrast, Assumption 2 means that any unused message is taken at face value, potentially providing a more attractive deviation. In the Supplementary Appendix, I provide an example of an IE assumed away

by Assumption 2 that differs from the structure of the IEs studied in this paper. In it, the threshold partitioning the state space is not equal to the sender’s expected utility from lying in equilibrium.

Discussion of the Model. This paper studies a model of cheap talk communication with three additional features: (i.) the receiver has access to a verification technology, (ii.) the verification technology reveals whether the sender’s message is true but not the state, and (iii.) the sender does not know the receiver’s verification cost when she chooses her message.

Communication Between a Politician and a Voter

This model applies naturally to communication between a politician, who wants to avoid blame for a policy failure, and a voter, who wants to accurately assign blame to the politician, when (i.) journalists will issue fact-checking reports of the politician’s statement, which the voter can access, (ii.) the fact-checking reports assess whether the politicians’ statement is true but not her culpability for the policy failure, and (iii.) the politician does not know which journalists will fact-check her statement, which determines how costly it is for the voter to verify.

In this application, β , the parameter in the receiver’s cost of verification that scales the realization κ , represents features of the political or media environment that affect the cost to access fact-checked information. For instance, larger values of β might represent a situation where few media organizations fact-check politicians’ statements. Or, larger values of β might represent a situation where the politician is unlikely to be fact-checked because she is a local politician and there are fewer journalists covering her.

Informational Lobbying

This model also applies to informational lobbying of a legislator by a non-allied lobbyist. Suppose a lobbyist wants to convince a legislator to support an energy policy, and the legislator wants to support the policy if and only if it will be successful. As part of the lobbyist’s efforts, she informs the legislator that studies have shown similar policies to be

successful when implemented elsewhere. In this example, (*i.*) the legislator can verify whether the lobbyist's message is true by finding and reading the studies, (*ii.*) this will reveal whether the lobbyist told the truth but not whether the policy will be successful, and (*iii.*) the lobbyist does not know the legislator's cost to verify because she does not know the legislator's time and resource constraints.

In this application, β might vary with the legislator's resources. For instance, given the difference in staff sizes, β might be larger for a member of the House of Representatives than a member of the Senate.

4 Analysis

4.1 Equilibrium Structure

I begin by considering the receiver's action as a function of the message he observes and any additional information learned from verification. When he chooses his action, he solves:

$$\max_{a \in [0,1]} -\mathbb{E}[(\theta - a)^2 | (m, v)].$$

This expression is uniquely maximized when $a(m, v) = \mathbb{E}[\theta | (m, v)]$.

Before choosing his action, the receiver decides whether to verify the sender's message. The optimality of $a^*(m, v)$ and the fact that the receiver has quadratic loss utility means he will verify the sender's message if:

$$\begin{aligned} & Var(\theta | m, c = 0) - \phi(m) Var(\theta | m, c = 1, v = t) - (1 - \phi(m)) Var(\theta | m, c = 1, v = f) \\ &= \underbrace{Var(\theta | m, c = 0) - (1 - \phi(m)) Var(\theta | m, c = 1, v = f)}_{\text{Informational Effect of Verification: } \Lambda(m)} > \beta \kappa, \end{aligned} \tag{1}$$

where $Var(\theta | \cdot)$ is the variance of the receiver's belief, $\phi(m)$ is the receiver's conjecture about the probability message m is sent truthfully, and the second line follows from the fact that

$Var(\theta|m, c = 1, v = t) = 0$. The left-hand side of (1) represents the informational effect of verification, which is the difference between the variance in the receiver's belief if he does not verify and his expected variance if he does. I denote this $\Lambda(m)$. From (1), it is clear the receiver will neither verify a message he believes is only sent truthfully, nor will he verify a message he believes is only sent as a lie since $\Lambda(m) = 0$ in both cases.

In any IE, some types of senders must lie. To see why, suppose this is not the case. Then, the receiver believes all of the sender's messages without verifying them. But then a low-type sender has a profitable deviation to sending a higher message. Hence, in any IE, at least one type of sender lies and receives an expected payoff u^{lie} , which must be independent of θ by Assumption 1.

In the following proposition, I provide a characterization result for any IE of this model.

Proposition 1. *In any IE, the sender tells the truth if $\theta > u^{lie}$ and lies if $\theta < u^{lie}$; when the sender lies, she lies by randomizing over each message $m > u^{lie}$ with a positive probability; and the receiver verifies all messages $m > u^{lie}$ with a positive probability.*

Proposition 1 states that any IE has a threshold structure pinned down by the sender's expected utility from lying. She tells the truth when the state exceeds this threshold and lies by randomizing over the messages above this threshold when the state is below it. As such $\Lambda(m) > 0$ for all m exceeding u^{lie} , so the receiver verifies all such messages with a positive probability.⁶ Moreover, except in the knife-edge case where $\theta = u^{lie}$, any message that is sent truthfully is sent as a lie and verified with a positive probability.

The intuition for this structure is as follows. As discussed above, in any IE, at least one type of sender must lie. Moreover, all messages $m \neq u^{lie}$ that are sent truthfully are also sent as lies.⁷ Suppose not. Then, there is a message m that is only sent truthfully. The receiver will not verify m and $a^*(m, \emptyset) = m$. Either $m > u^{lie}$ or $m < u^{lie}$. The sender can

⁶Proposition 1 provides a characterization, not existence. Without more structure on Θ , it is difficult to derive a necessary and sufficient condition for existence. However, in Section 4.2, I solve for all IE for a particular Θ . Moreover, in the Supplementary Appendix, I provide sufficient conditions for the existence of some IEs.

⁷In the Appendix, I address the knife-edge case where $\theta = u^{lie}$.

deviate from lying to sending m in the former. In the latter, the sender can deviate from telling the truth with m to lying.

Consider a sender with type $\theta > u^{lie}$ and suppose she lies. As a result, her expected utility is u^{lie} . Since the type- θ sender lies, $m = \theta$ must be off the equilibrium path. This is because Assumption 1 restricts the sender's strategy when lying to be independent of θ , meaning $m = \theta$ cannot be sent as a lie. Since $m = \theta$ is off the equilibrium path, Assumption 2 implies the sender can deviate from lying to reporting $m = \theta$, which will be believed. Thus, a sender of type $\theta > u^{lie}$ tells the truth.

Now consider a sender with type $\theta < u^{lie}$, and suppose she tells the truth. When the receiver verifies $m = \theta$, he chooses $a^*(m, t) = \theta < u^{lie}$. As discussed above, it must be that $m = \theta$ is also sent as a lie. This implies the action the receiver takes if he does not verify $m = \theta$ is also smaller than u^{lie} since the sender tells the truth if $\theta > u^{lie}$. Hence, the sender with type $\theta < u^{lie}$ who tells the truth has a profitable deviation to lying.

Finally consider the knife-edge case where $\theta = u^{lie}$; in this case, either $m = \theta$ is only sent truthfully or it is off the equilibrium path. Suppose this is not the case. Then $m = \theta$ is sent truthfully and as a lie since Assumption 1 means that the sender of type $\theta = u^{lie}$ cannot lie. When the receiver verifies this message, which happens with a positive probability since $\Lambda(m) > 0$, she chooses $a^*(m, f) < u^{lie}$ as the sender lies only when $\theta < u^{lie}$. But when she does not verify, she chooses $a^*(m, \emptyset) \leq u^{lie}$ since the only types of senders who send $m = \theta$ have types weakly below u^{lie} . Thus, it cannot be that $\theta = u^{lie}$.

As this discussion reveals, Assumptions 1 and 2 play critical roles in the fact that the threshold in the sender's strategy equals u^{lie} . Together, they imply that if a sender with type $\theta > u^{lie}$ lies, there is a profitable deviation to $m = \theta$. Relaxing either assumption disrupts this logic. Without Assumption 1, other types may lie with $m = \theta$, placing it on the equilibrium path; when the receiver sees such a message, she no longer believes it. As such, it is possible to construct an equilibrium where a sender of type $\theta > u^{lie}$ lies. Without Assumption 2, the off-path message $m = \theta$ may be met with a punishing belief rendering it

an unattractive deviation. This allows for the construction of an equilibrium where a sender of type $\theta > u^{lie}$ lies. In the Supplementary Appendix, I provide examples of both types of equilibria.

More generally, Assumptions 1 and 2 select equilibria from a larger set of equilibria that include equilibria where types $\theta > u^{lie}$ lie and equilibria where some on the equilibrium path messages are only sent as lies, as the examples in the Supplementary Appendix illustrate. Notably, however, truth-only on the equilibrium path messages remain impossible regardless of either assumption, since a message sent only truthfully will be believed, yielding a payoff above u^{lie} if $m > u^{lie}$ or a payoff below u^{lie} if $m < u^{lie}$.

4.2 Three-State Example

Proposition 1 characterizes the structure of any IE of this model. In this section, I focus on the case where $\theta \in \Theta = \{0, \theta_2, 1\}$ with $\theta_2 \in (0, 1)$, $h(\theta) = \frac{1}{3}$ for all θ , and $\kappa \sim \mathcal{U}[0, 1]$.⁸ This case maps well into the example of informational lobbying about a policy that may have a negative ($\theta = 0$), middling ($\theta = \theta_2$) or positive impact ($\theta = 1$) on a politician's—the receiver's—district. A lobbyist—the sender—who wants the politician to support the policy, communicates with the politician about studies that show the policy's impact when implemented elsewhere. After observing the lobbyist's message, the politician can verify it by directing his staffer to research whether the studies the lobbyist cited support the claim she made.

The following proposition establishes all IEs given these preliminaries.

Proposition 2. *Define*

$$\tilde{\beta} \equiv \frac{(2 - \theta_2)^3}{36(1 - 2\theta_2)} \text{ and } \bar{\beta} \equiv \frac{1}{4(1 - 2\theta_2)}.$$

⁸Note, in the traditional cheap talk model with these preliminaries, there is no influential equilibrium. To see this, suppose not. Then there must be m and m' such that $a^*(m) \neq a^*(m')$. But then $a^*(m) > a^*(m')$ or $a^*(m) < a^*(m')$, which implies there is a profitable deviation.

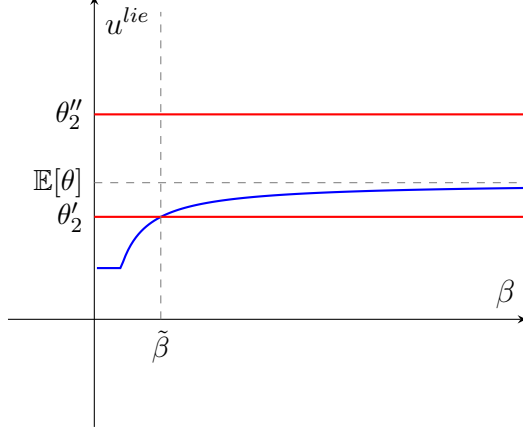


Figure 1: Expected utility from lying with $m = 1$ and from deviating to $m = \theta_2$, which is off the equilibrium path, for the sender of types $\theta \in \{0, \theta_2\}$. Assumes $\theta_2' = \frac{1}{3}$ and $\theta_2'' = \frac{2}{3}$.

An IE exists in all regions of the parameter space. Moreover, any IE is of one of the following:

When $\theta_2 < \frac{1}{2}$

- (a.) and $\beta \geq \tilde{\beta}$, an IE exists where the sender tells the truth when $\theta = 1$ and lies when $\theta \in \{0, \theta_2\}$ by sending $m = 1$;*
- (b.) and $\beta \leq \tilde{\beta}$, where $\tilde{\beta} < \bar{\beta}$, at least one IE exists where the sender tells the truth when $\theta \in \{\theta_2, 1\}$ and lies when $\theta = 0$ by randomizing between sending $m = 1$ with probability σ^* and $m = \theta_2$ with probability $1 - \sigma^*$.*

And when $\theta_2 \geq \frac{1}{2}$,

- (c.) at least one IE exists where the sender tells the truth when $\theta \in \{\theta_2, 1\}$ and lies when $\theta = 0$ by randomizing between sending $m = 1$ with probability σ^* and $m = \theta_2$ with probability $1 - \sigma^*$.*

Figure 1 provides a visual intuition for Proposition 2(a) by depicting the type-0 and type- θ_2 senders' expected utility from lying with $m = 1$ as a function of β .⁹ This expected utility is increasing in β since the probability the receiver verifies $m = 1$ is decreasing in β , and a lower probability of verification means a higher probability of successfully pooling

⁹Recall, the receiver pays a cost $\beta\kappa$ to verify the sender's message, where $\kappa \sim \mathcal{U}[0, 1]$.

with the type-1 sender. For this type of IE to exist, this expected utility must be weakly larger than θ_2 , which is the payoff that can be obtained by deviating to $m = \theta_2$, which is off the equilibrium path. When $\theta_2 < \frac{1}{2}$ (e.g., $\theta_2 = \theta'_2$ in Figure 1) and β is sufficiently large—larger than $\tilde{\beta}$ —this type of IE exists. Additionally, as β approaches infinity, the probability of verification approaches zero, and u^{lie} approaches the payoff the sender achieves in the traditional cheap talk model with these preliminaries, $\mathbb{E}[\theta]$. However, when $\theta_2 \geq \frac{1}{2}$ (e.g., $\theta_2 = \theta''_2$ in Figure 1), this type of IE does not exist.

Figure 2 provides a visual intuition for Proposition 2(b) and (c). It depicts the expected utility of the type-0 sender when she lies with $m = 1$ and $m = \theta_2$ as a function of σ , the probability she lies with $m = 1$. The type-0 sender's expected utility from lying with $m = 1$ is weakly decreasing in σ since as σ increases, the receiver is more likely to verify the message and takes a lower action when he does not.¹⁰ On the other hand, as σ increases, the receiver is less likely to verify $m = \theta_2$ and takes a higher action when he does not. So the type-0 sender's expected utility from lying with $m = \theta_2$ is weakly increasing in σ .¹¹

To complete the intuition for Proposition 2(b) and (c), there are two cases to consider. First, suppose $\theta_2 < \frac{1}{2}$. Let $\underline{\beta}$ solve

$$\frac{1 - \sqrt{1 - 4\underline{\beta}} - 2\underline{\beta}}{2\underline{\beta}} = \frac{4\underline{\beta} - \theta_2^2 + \sqrt{-4\underline{\beta}\theta_2^2 + \theta_2^4}}{2\underline{\beta}}.$$

If $\beta \in (\underline{\beta}, \bar{\beta})$, the curves depicted in the left panel of Figure 2 have a unique intersection point, denoted σ^* .¹² When $\beta \geq \bar{\beta}$, lying with $m = 1$ becomes too attractive to the type-0 sender, and there is no longer a σ^* such that she is indifferent between lying with $m = 1$ and $m = \theta_2$. And when $\beta \leq \underline{\beta}$, there are a continuum of σ^* such that she is indifferent between lying with $m = 1$ and $m = \theta_2$ because the receiver verifies both messages with certainty. This case is depicted in the right panel of Figure 2.

¹⁰In fact, as depicted in Figure 2, for β sufficiently small and σ sufficiently large ($\sigma > \sigma_1(\beta)$), the type-0 sender's expected utility from lying with $m = 1$ is zero since the receiver verifies the message with certainty.

¹¹Although not depicted in Figure 2, for β sufficiently small and σ sufficiently small, the sender's expected utility from lying with $m = \frac{1}{2}$ is zero since the receiver verifies the message with certainty.

¹²For more information, see the proof of Proposition 2 in the Appendix.

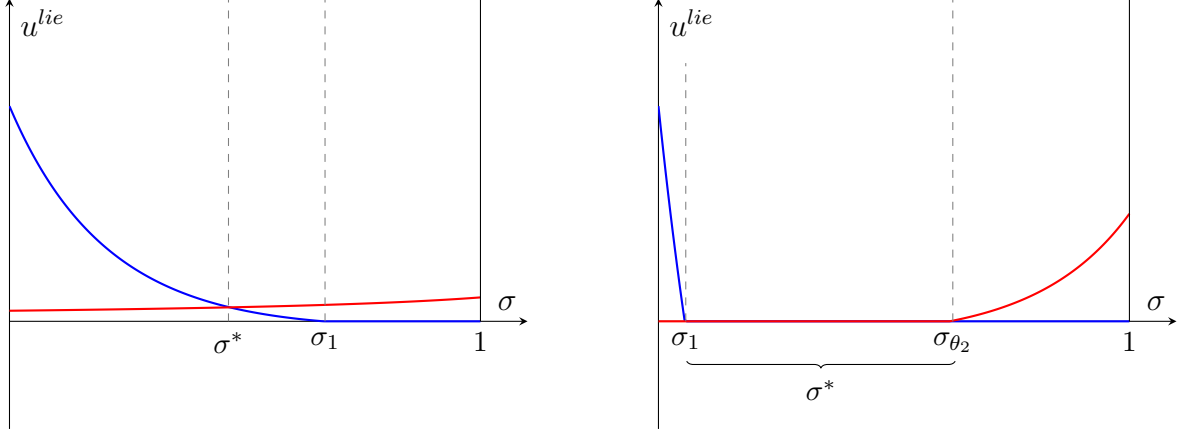


Figure 2: Left panel: Expected utility from lying with $m = 1$ and $m = \theta_2$ for a sender of type $\theta = 0$. Assumes $\beta = \frac{6}{25}$ and $\theta_2 = \frac{1}{3}$. Right panel: Expected utility from lying with $m = 1$ and $m = \theta_2$ for a sender of type $\theta = 0$. Assumes $\beta = \frac{1}{20}$ and $\theta_2 = \frac{1}{3}$.

Now suppose $\theta_2 \geq \frac{1}{2}$. Then lying with $m = 1$ is never too attractive to the type-0 sender that she cannot be made indifferent between lying with $m = 1$ and $m = \theta_2$. Hence, there is no $\bar{\beta}$ such that this type of IE does not exist.

In general, every message in the support of the sender's strategy is sent truthfully and as a lie, and, as such, is verified with a positive probability. However, this insight breaks down for the knife-edge case $\beta = \bar{\beta}$. When this is the case, an IE exists where the sender tells the truth when $\theta \in \{\theta_2, 1\}$ and lies when $\theta = 0$ by sending $m = 1$. Thus, $m = \theta_2$ is only sent truthfully and, as a result, is never verified. The key to understanding this IE is that in it $u^{lie} = \theta_2$. Hence, neither the type- θ_2 nor the type-0 sender have a profitable deviation.¹³

One might imagine that as β approaches zero, there will be an IE in which the sender always tells the truth. As discussed above, this cannot be the case; there cannot be an IE where the sender never lies. This is due to the endogenous nature of verification. If the receiver believes the sender never lies, he has no reason to verify the sender's messages—even if the cost of doing so approaches zero. But then, the sender has the ability to lie without fear of being caught. This insight illustrates how the existence of influential communication

¹³This particular IE has a similar logic to equilibrium characterization in [Sadakane and Tam \(2026\)](#), where the sender sends a vague intermediate message that is believed and yields the same expected utility to the sender as lying with higher messages.

between a biased sender and receiver requires an accurate suspicion that the sender may be lying. Without this suspicion, the receiver never verifies the sender's message, but then the sender can lie freely, destroying influential communication.

The following proposition shows what happens when β approaches infinity.

Proposition 3. *When $\beta \rightarrow \infty$, $\mathbb{E}[\theta|(m, \emptyset)] \rightarrow \mathbb{E}[\theta]$ for all m in the support of the sender's strategy.*

When the probability the receiver verifies the sender's messages approaches zero, the model approaches traditional cheap talk in which in equilibrium the receiver's belief on the equilibrium path is equal to the prior. The precise structure of the limit equilibrium depends on θ_2 . When $\theta_2 < \frac{1}{2}$, the type-0 and type- θ_2 sender lie by sending $m = 1$.¹⁴ As a result, when the receiver does not verify the highest message, his belief is equivalent to the prior. When $\theta_2 \geq \frac{1}{2}$, the type-0 sender lies by randomizing between $m = \theta_2$ and $m = 1$. In equilibrium, she must be indifferent between these lies. In particular, in the limit, this means she must be indifferent between the two actions the receiver takes when he does not verify.

The Effects of Verification Let Γ^{CT} denote the version of the three-state model that is identical except the receiver does not have the ability to verify the sender's message. Put differently, Γ^{CT} is the traditional cheap talk model. Perhaps unsurprisingly, the ability to verify the sender's message improves the receiver's *ex ante* expected utility relative to his *ex ante* expected utility in Γ^{CT} .¹⁵

Proposition 4. *In any IE, the receiver's ex ante expected utility is strictly higher than his ex ante expected utility in Γ^{CT} .*

There are two reasons the receiver's *ex ante* expected utility is higher when he can verify the sender's message than in Γ^{CT} . The first is that verification has a direct *informational* effect, $\Lambda(m)$, which is the difference between the variance in the receiver's

¹⁴In fact, this is true for any β such that this IE exists.

¹⁵In Γ^{CT} , no informative equilibria exists.

belief if he does not verify and his expected variance if he does. Although verification is costly, the receiver chooses to verify when this informational effect exceeds this cost, and when $\beta\kappa > \Lambda(m)$, he strictly prefers to verify.¹⁶

Verification also increases the receiver's expected utility through an indirect *deterrence* effect. In equilibrium, verification generates an endogenous cost of lying for the type- θ_2 sender. If she lies and is caught, she will be pooled with the type-0 sender when the receiver chooses $a^*(m, f)$. In some cases, this cost of lying deters the type- θ_2 sender from lying.

The deterrence effect can be seen in the IE described by Proposition 2(b) and (c).¹⁷ In this IE, the type- θ_2 sender tells the truth. Deviating to lying with $m = 1$ leads to a payoff of $a^*(1, \emptyset) > a^*(\theta_2, \emptyset)$ if the receiver does not verify. But when he does, the type- θ_2 sender is punished by the receiver choosing an action that pools her with the type-0 sender who also lies. In this IE, this punishment is costly enough to deter the type- θ_2 sender from lying.

While the informational effect of verification is present in all IEs, the deterrence effect is not—in the IE described by Proposition 2(a), the type- θ_2 sender is not deterred from lying. Moreover, when $\theta_2 < \frac{1}{2}$ and $\beta \in (\tilde{\beta}, \bar{\beta})$, two IEs exist: one where verification has a deterrence effect and an informational effect and one where verification only has an informational effect. A natural conjecture is that these effects operate like substitutes: if verification deters lying, it reduces the informational value of verifying the sender's message. In the following proposition, I show this is not necessarily the case. To do so, I first introduce the concept of the expected informational effect of verification:

$$\mathbb{E}_{m \in M^*}[\Lambda(m)] = \sum_{m \in M^*} \rho(m)(\text{Var}(\theta|m, c = 0, v = \emptyset) - (1 - \phi(m))(\text{Var}(\theta|m, c = 1, v = f)))$$

where $\rho(m)$ is the probability m is sent on the equilibrium path and M^* is the set of messages

¹⁶In contrast, in [Sadakane and Tam \(2026\)](#), the direct informational effect is completely offset by the cost of verification because the equilibrium is constructed such that when the receiver verifies, he is exactly indifferent between verifying and not verifying.

¹⁷This IE exists simultaneously with the IE in which verification only provides an informational effect, described by Proposition 2(a), when $\theta_2 < \frac{1}{2}$ and $\beta \in (\tilde{\beta}, \bar{\beta})$

in the support of the sender's strategy.¹⁸

Proposition 5. *Suppose $\theta_2 \in (2 - \sqrt{3}, \frac{1}{2})$ and $\beta \in (\tilde{\beta}, \bar{\beta})$. For β sufficiently close to $\bar{\beta}$, the expected informational effect of verification is greater in the IE where there is a deterrence effect than in the IE where there is not.*

An intuition for this result is as follows. In the IE without deterrence, the sender lies when $\theta \in \{0, \theta_2\}$ and tells the truth when $\theta = 1$. As such, when the receiver verifies $m = 1$ and learns the message was a lie, there is residual uncertainty—he is unsure whether the state is 0 or θ_2 . By contrast, in the IE with deterrence, only the type-0 sender lies—when the sender verifies a message, he learns the state. Hence, there is a sense in which deterrence can make verification more effective.

In particular, when θ_2 is sufficiently large, which minimizes the informational effect of verification in the IE without deterrence and when $\sigma^* \rightarrow 1$, which is the case when $\beta \rightarrow \bar{\beta}$, and which maximizes the informational effect of verification in the IE with deterrence, the expected informational effect in the latter IE is larger than the former.

The connection between the informational effect of verification and probability of verification yields the following corollary.

Corollary 1. *Suppose $\theta_2 \in (2 - \sqrt{3}, \frac{1}{2})$, and $\beta \in (\tilde{\beta}, \bar{\beta})$. For β sufficiently close to $\bar{\beta}$, the ex-ante probability of verification is greater in the IE where there is a deterrence effect than in the IE where there is not.*

5 Conclusion

In this paper, I study cheap talk communication between a biased sender and a receiver who can verify the sender's message by paying a privately known cost. This setting applies to many strategic interactions in politics, such as a lobbyist communicating with a politician

¹⁸Recall, $\phi(m)$ is the probability m is sent truthfully, c denotes the receiver's decision, v denotes whether the sender's message is true, false, or is not checked, and $\Lambda(m)$ is the informational effect of verification.

who can investigate the lobbyist’s claims or a politician who communicates with voters who can turn to fact-checkers or journalists to verify the politician’s claims.

I show that any influential equilibrium of the model is characterized by a threshold structure pinned down by the sender’s expected utility from lying in equilibrium. When the state exceeds this threshold, the sender tells the truth; when the state is below this threshold, she lies by mixing over all messages that exceed the threshold. This means that all messages sent truthfully are sent as lies—except in a knife-edge case—and the receiver verifies all such messages with a positive probability.

This equilibrium characterization reveals that there is dependence between the success of endogenous verification in inducing informative communication and the suspicion of lying. If the receiver believes there is a message that is only sent truthfully, he has no incentive to verify it. But if there is a message that is believed and not verified, the sender must have a profitable deviation (except in the knife-edge case where the sender’s expected utility from lying equals one of the states of the world): either the expected utility of lying is below that message—in which case there is a profitable deviation from lying to sending that message—or the expected utility of lying is above that message—in which case where there is a profitable deviation from telling the truth with that message to lying. Thus, there are no messages that are only sent truthfully. This insight differs from existing work on cheap talk with verification ([Sadakane and Tam, 2026](#); [Dziuda and Salas, 2018](#)).

I then analyze the model when there are three equally likely states, and the cost to verify is drawn from a uniform distribution. This analysis illustrates that verification has two effects: an informational effect and a deterrence effect. I explore these effects and show that, in some cases, the presence of the deterrence effect increases the expected informational effect.

One natural extension would be to study what happens when there are multiple senders, one whose utility is increasing in the sender’s action, and one whose utility is decreasing. For example, imagine a politician who is lobbied by opposing lobbyists. I leave this extension

for future research.

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6 Appendix: Main Results

6.1 Proof of Proposition 1

Proof. Suppose an IE exists.

- (i.) Claim: At least one message is sent as a lie.

Proof. Suppose not. Then all messages are sent truthfully. When the receiver observes a message, he believes it, will not verify, and will choose $a^*(m, \emptyset) = m$. But then a sender of type $\theta < \theta_N$, where θ_N is the highest state, has a profitable deviation to $m = \theta_N$. ■

- (ii.) Claim: At least one message is sent truthfully.

Proof. Suppose not. Then all messages are sent as lies and the receiver never verifies. By Assumption 1, the sender's lying strategy cannot depend on her type, which means $\mathbb{E}[\theta|m]$ is constant for all m . But then $a^*(m, v)$ is constant on the equilibrium path. This violates the definition of an IE. ■

- (iii.) Claim: The sender's expected utility from lying with message m is constant for all lies in the support of the sender's strategy.

Proof. Let M_f be the set of messages that are sent as lies on the equilibrium path. Suppose the claim is false. Then there exist lies $m \in M_f$ and $m' \in M_f$ such that the sender's expected utility from lying with m is strictly higher than lying with m' . But then a liar who was supposed to lie with m' could deviate to lying with m . Hence, this is not an IE. ■

- (iv.) Claim: Each message m such that $m \neq u^{lie}$ and m is in the support of the sender's strategy is sent as a lie.

Proof. Let M_t be the set of messages that are sent truthfully on the equilibrium path, and let $M^* = M_f \cup M_t$ be the set of messages in the support of the sender's strategy. Suppose the claim is false. Then there exists some $m \neq u^{lie} \in M^* \setminus M_f$, where u^{lie} is the sender's expected utility from lying. When the receiver observes m , he will not verify and will choose $a^*(m, \emptyset) = m$.

Suppose $m > u^{lie}$. Then a liar can deviate from lying according to her equilibrium strategy to lying by reporting m . The message will be believed since $m \in M^* \setminus M_f$ by Assumption 2, so it is a profitable deviation.

Suppose $m < u^{lie}$. Then the type- m sender can deviate from truthfully reporting m , in which case she receives a payoff of m , to lying and get u^{lie} . This is a profitable deviation. ■

- (v.) Claim: For any θ such that $\theta \neq u^{lie}$, the sender either lies with probability one or tells the truth with probability one.

Proof. Suppose not. Consider the type- θ where $\theta \neq u^{lie}$ who randomizes between lying and telling the truth. She must be indifferent between telling the truth and lying. By Assumption 1, all senders who lie must use the same strategy, so $m = \theta$ can only be sent truthfully. Hence, when the sender sends m , the receiver does not verify and chooses $a^*(m, \emptyset) = m$. It is immediate that the sender cannot be indifferent between telling the truth and lying. ■

- (vi.) Claim: For any θ such that $\theta > u^{lie}$, the sender tells the truth with probability one.

Proof. Suppose not. Then there exists a θ such that $\theta > u^{lie}$ and the type- θ sender lies. Then $m = \theta$ is not in the support of the sender's strategy since the sender's strategy when lying is independent of θ and the type- θ cannot lie with $m = \theta$. Since m is off the equilibrium path, when the sender lies and receives u^{lie} , she can deviate to

the message m . This message will be believed by Assumption 2, and the receiver will choose $a^*(m, \emptyset) = \theta$. By assumption, this is larger than u^{lie} . ■

(vii.) Claim: For any θ such that $\theta < u^{lie}$, the sender lies with probability one.

Proof. Suppose not. Then there exists θ such that $\theta < u^{lie}$ and the type- θ sender tells the truth.

The type- θ sender's expected utility from telling the truth is $\pi^*(m)\theta + (1 - \pi^*(m))a^*(m, \emptyset)$, where $\pi^*(m)$ is the probability the receiver verifies the message m in equilibrium. Optimality implies $\pi^*(m)\theta + (1 - \pi^*(m))a^*(m, \emptyset) \geq u^{lie}$, which requires $a^*(m, \emptyset) > u^{lie}$ since $\theta < u^{lie}$. However, (vi.) shows that the sender tells the truth if $\theta > u^{lie}$. Hence $a^*(m, \emptyset) \leq \mathbb{E}[\theta | \theta \leq u^{lie}] < u^{lie}$. This is a contradiction. ■

(viii.) Claim: If there exists a θ such that $\theta = u^{lie}$, $m = \theta$ is only sent truthfully or is off the equilibrium path.

Proof. Suppose, toward a contradiction, that $\theta = u^{lie}$ and $m = \theta$ is sent as a lie. Then, by Assumption 1, $m = \theta$ must also be sent truthfully. By (vi.), no type above u^{lie} lies. Therefore, $a^*(m, f) < u^{lie}$.

Since $m = \theta$ is sent as a lie, the sender's expected utility from lying with this message is

$$\pi^*(m)a^*(m, f) + (1 - \pi^*(m))a^*(m, \emptyset).$$

By (iii.), this must equal u^{lie} .

But since $a^*(m, f) < u^{lie}$, equality requires $a^*(m, \emptyset) > u^{lie}$. However, this is impossible since $a^*(m, \emptyset) \leq u^{lie}$ due to only types $\theta \leq u^{lie}$ since $m = \theta$. Therefore lying with $m = \theta$ yields strictly less than u^{lie} , contradicting (iii). Hence $m = \theta$ is not sent as a lie. ■

■

6.2 Proof of Proposition 2

Let $\pi^*(m)$ denote the probability message m is verified in equilibrium.

Proof. Proposition 1 implies that in the three-state example, there are three cases to consider:

$u^{lie} > \theta_2$, $u^{lie} < \theta_2$, and $u^{lie} = \theta_2$.

$u^{lie} > \theta_2$:

Suppose an IE exists where $u^{lie} > \theta_2$. Then the sender lies if $\theta \in \{0, \theta_2\}$ and tells the truth otherwise. Since the sender lies by randomizing over all $m > u^{lie}$, due to Proposition 1, the only message used by types 0 and θ_2 when lying is $m = 1$.

A sender of type $\theta \in \{0, \theta_2\}$ has the following expected utility from lying:

$$\min \left\{ \frac{(2 - \theta_2)^2}{18\beta}, 1 \right\} u_S \left(\frac{\theta_2}{2} \right) + \max \left\{ 1 - \frac{(2 - \theta_2)^2}{18\beta}, 0 \right\} u_S \left(\frac{1 + \theta_2}{3} \right), \quad (2)$$

where $\pi^*(1) = \min \left\{ \frac{(2 - \theta_2)^2}{18\beta}, 1 \right\}$. This expected utility is strictly increasing in β for $\beta > \frac{(2 - \theta_2)^2}{18}$, which implies $\pi^*(1) \in (0, 1)$, and is constant otherwise.

The type-1 sender has the following expected utility from telling the truth:

$$\min \left\{ \frac{(2 - \theta_2)^2}{18\beta}, 1 \right\} u_S(1) + \max \left\{ 1 - \frac{(2 - \theta_2)^2}{18\beta}, 0 \right\} u_S \left(\frac{1 + \theta_2}{3} \right). \quad (3)$$

It remains to check whether any type of sender has a profitable deviation. A sender of type $\theta \in \{0, \theta_2\}$ lacks a profitable deviation to $m = \theta_2$, which will be believed by Assumption 2, if and only if $(2) \geq \theta_2$. Moreover, since $(3) > (2)$, $(2) \geq \theta_2$ implies the type-1 sender will not deviate to $m = \theta_2$. Thus, the necessary and sufficient condition for existence of this IE is

$$(2) \geq \theta_2 \quad (4)$$

If $\beta \leq \frac{(2-\theta_2)^2}{18}$ or if $\theta_2 \geq \frac{1}{2}$, (4) cannot be satisfied. So suppose $\beta > \frac{(2-\theta_2)^2}{18}$ and $\theta_2 < \frac{1}{2}$. Solving (4) yields $\theta \geq \tilde{\beta}$, where

$$\tilde{\beta} = \frac{(2 - \theta_2)^3}{36(1 - 2\theta_2)}, \quad (5)$$

with $\tilde{\beta} > 0$ since $\theta_2 < \frac{1}{2}$. Moreover,

$$\tilde{\beta} > \frac{(2 - \theta_2)^2}{18},$$

so $\beta \geq \tilde{\beta}$ implies $\beta > \frac{(2-\theta_2)^2}{18}$. Hence this IE exists if and only if $\theta_2 < \frac{1}{2}$ and $\beta \geq \tilde{\beta}$. Additionally, note that $\beta = \bar{\beta}$ is the knife-edge case where $\theta_2 = u^{lie}$ and $m = \theta_2$ is off the equilibrium path.

$u^{lie} < \theta_2$:

Suppose an IE exists where $u^{lie} < \theta_2$. Then types θ_2 and 1 tell the truth and type 0 lies by randomizing over $m \in \{\theta_2, 1\}$. Let σ^* denote the probability that type 0 lies with $m = 1$ in such an IE. In equilibrium, σ^* must satisfy

$$\max \left\{ 1 - \frac{\sigma}{\beta(1 + \sigma)^2}, 0 \right\} u_S \left(\frac{1}{1 + \sigma} \right) = \max \left\{ 1 - \frac{(1 - \sigma)\theta_2^2}{\beta(2 - \sigma)^2}, 0 \right\} u_S \left(\frac{\theta_2}{2 - \sigma} \right), \quad (6)$$

where the LHS is the type-0 sender's expected utility from lying with $m = 1$ and the RHS is his expected utility from lying with $m = \theta_2$.

The LHS is weakly decreasing in σ while the RHS is weakly increasing. Hence, by continuity, a solution exists if and only if $\max \left\{ \frac{4\beta-1}{8\beta}, 0 \right\} < \theta_2$. If $\theta_2 < \frac{1}{2}$, this holds if and only if $\beta < \bar{\beta}$, where

$$\bar{\beta} = \frac{1}{4(1 - 2\theta_2)}. \quad (7)$$

If $\theta_2 \geq \frac{1}{2}$, it holds for all $\beta > 0$.

It remains to consider whether any type of sender has a profitable deviation. Since all sender types receive a positive payoff in equilibrium, none have an incentive to deviate to $m = 0$, which is off the equilibrium path and believe by Assumption 2. Moreover since when the sender tells the truth the receiver verifies the truthfulness of her message with a positive probability and $\theta_2 > u^{lie}$, neither type θ_2 nor type 1 have a profitable deviation to lying.

$u^{lie} = \theta_2$:

Finally, suppose an IE exists where $u^{lie} = \theta_2$ and $m = \theta_2$ is only sent truthfully (the other knife-edge case was previously discussed). Thus, in equilibrium, the type-0 sender lies with $m = 1$. Otherwise, the sender tells the truth. Since $m = \theta_2$ is only sent as a lie, the type-0 sender's payoff is

$$\frac{4\beta - 1}{8\beta}.$$

So, $u^{lie} = \theta_2$ if and only if $\beta = \bar{\beta}$.

Combining cases, if $\theta_2 < \frac{1}{2}$, an IE in which types $\{0, \theta_2\}$ lie exists if and only if $\beta \geq \tilde{\beta}$, while an IE in which only type 0 lies exists if and only if $\beta \leq \bar{\beta}$. Furthermore, $\tilde{\beta} < \bar{\beta}$. If $\theta_2 \geq \frac{1}{2}$, the former IE does not exist and the latter exists for all $\beta > 0$.

■

6.3 Proof of Proposition A

Proposition A. Define $\underline{\beta}$ as the unique solution to

$$\frac{1 - \sqrt{1 - 4\underline{\beta}} - 2\underline{\beta}}{2\underline{\beta}} = \frac{4\underline{\beta} - \theta_2^2 + \sqrt{-4\underline{\beta}\theta_2^2 + \theta_2^4}}{2\underline{\beta}}.$$

If $\beta < \underline{\beta}$, a continuum of IEs exist where the sender tells the truth when $\theta \in \{\theta_2, 1\}$ and lies when $\theta = 0$ by randomizing between $m = 1$ with probability σ^* and $m = \theta_2$ with probability $1 - \sigma^*$.

Proof. Consider the following strategy for the sender: the sender tells the truth when $\theta \in$

$\{\theta_2, 1\}$ and lies when $\theta = 0$ by randomizing between $m = 1$ with probability σ and $m = \theta_2$ with probability $1 - \sigma$.

Define $\sigma_1(\beta)$ and $\sigma_{\theta_2}(\beta)$ as the unique values of $\sigma \in [0, 1]$ satisfying $\pi^*(1) = 1$ and $\pi^*(\theta_2) = 1$ respectively:

$$\sigma_1(\beta) = \frac{1 - \sqrt{1 - 4\beta} - 2\beta}{2\beta},$$

$$\sigma_{\theta_2}(\beta) = \frac{4\beta - \theta_2^2 + \sqrt{\theta_2^4 - 4\beta\theta_2^2}}{2\beta}.$$

The receiver verifies $m = 1$ with certainty if and only if $\sigma \geq \sigma_1(\beta)$, and verifies $m = \theta_2$ with certainty if and only if $\sigma \leq \sigma_{\theta_2}(\beta)$.

By definition of $\underline{\beta}$, $\sigma_1(\underline{\beta}) = \sigma_{\theta_2}(\underline{\beta})$. Since σ_1 is increasing in β and σ_{θ_2} is decreasing in β , it follows that for $\beta < \underline{\beta}$, $\sigma_1(\beta) < \sigma_{\theta_2}(\beta)$, so the interval $[\sigma_1(\beta), \sigma_{\theta_2}(\beta)]$ is non-empty.

For any $\sigma \in [\sigma_1(\beta), \sigma_{\theta_2}(\beta)]$, the receiver verifies both $m = 1$ and $m = \theta_2$ with certainty. When either message is verified and found false, the receiver learns $\theta = 0$ and chooses $a = 0$. Hence the type-0 sender's expected utility from lying with either message is 0, so condition (6) is satisfied for every $\sigma \in [\sigma_1(\beta), \sigma_{\theta_2}(\beta)]$.

It remains to verify that no sender type has a profitable deviation. Since all sender types receive a weakly positive payoff, none will deviate to $m = 0$, which is off the equilibrium path, and yields $a^*(0, \emptyset) = 0$ by Assumption 2. The type- θ_2 sender receives θ_2 from telling the truth; deviating to $m = 1$ yields $a^*(1, f) = 0 < \theta_2$, since the receiver verifies with certainty and the message is false. The type-1 sender receives 1 from telling the truth; deviating to $m = \theta_2$ yields $a^*(\theta_2, f) = 0 < 1$. So there are no profitable deviations. ■

6.4 Proof of Proposition 3

Proof. If $\theta_2 \in (0, \frac{1}{2})$, then as $\beta \rightarrow \infty$ the unique IE is one where the sender tells the truth when $\theta = 1$ and lies when $\theta \in \{0, \theta_2\}$. In this IE, $\mathbb{E}[\theta|(1, \emptyset)] = \frac{\theta_2 + 1}{3} = \mathbb{E}[\theta]$.

If $\theta_2 \in [\frac{1}{2}, 1)$, then as $\beta \rightarrow \infty$ the unique IE is one where the sender tells the truth when $\theta \in \{\theta_2, 1\}$ and lies when $\theta = 0$ by randomizing between $m = 1$ and $m = \theta_2$. As $\beta \rightarrow \infty$, σ^* solves $\frac{1}{1+\sigma} = \frac{\theta_2}{2-\sigma} \Leftrightarrow \sigma^* = \frac{2-\theta_2}{1+\theta_2}$. Plugging σ^* in, $\mathbb{E}[\theta|(1, \emptyset)] = \mathbb{E}[\theta|(\theta_2, \emptyset)] = \frac{\theta_2+1}{3} = \mathbb{E}[\theta]$. ■

6.5 Proof of Proposition 4

Proof. In Γ^{CT} , the receiver's expected utility is:

$$-Var(\theta) = -\frac{2(1 - \theta_2 + \theta_2^2)}{9}. \quad (8)$$

Now consider any IE. Slightly abusing notation, let $u_R(m, \emptyset)$ denote the receiver's expected utility when he does not verify the sender's message. Given message m , the receiver chooses $a^*(m, \emptyset) = \mathbb{E}[\theta|m]$, so his expected utility is

$$u_R(m, \emptyset) = -Var(\theta|m).$$

Taking the expectation over messages and applying the law of total variance

$$\mathbb{E}[Var(\theta|m)] \leq Var(\theta),$$

with strict inequality in an IE. Hence, $u_R(m, \emptyset) > (8)$. Due to the optimality of the receiver's best response, the fact that verification reduces variance, and the fact that receiver strictly prefers to verify for some realizations of κ , this is sufficient. ■

6.6 Proof of Proposition 5

To prove Proposition 5, I begin with the following lemma.

Lemma A. *Consider the IE described by Proposition 2(b). If $\beta \in (\underline{\beta}, \overline{\beta})$, $\sigma^*(\beta)$ is continuous.*

Proof. Define an implicit function $F(\sigma, \beta) = u_S^1(\sigma, \beta) - u_S^{\theta_2}(\sigma, \beta, \theta_2)$, where $u_S^1(\sigma, \beta)$ and $u_S^{\theta_2}(\sigma, \beta, \theta_2)$ are the type-0 sender's expected utility from lying with $m = 1$ and $m = \theta_2$ respectively. Proposition 2 implies that for every $\beta \in (\underline{\beta}, \bar{\beta})$ there exists a unique $\sigma^*(\beta)$ such that $F(\sigma^*(\beta), \beta) = 0$. Proposition 2 also shows that $u_S^1(\sigma, \beta)$ and $u_S^{\theta_2}(\sigma, \beta)$ are continuously differentiable in β and σ . Hence, $F(\sigma, \beta)$ is too. Applying the implicit function theorem, for an arbitrary $\beta \in (\underline{\beta}, \bar{\beta})$, there exists a unique, differentiable function $\omega(\beta)$ such that in the neighborhood of β , $F(\omega(\beta), \beta) = 0$ as long as $\frac{\partial u_S^1}{\partial \sigma}(\omega(\beta), \beta) - \frac{\partial u_S^{\theta_2}}{\partial \sigma}(\omega(\beta), \beta) \neq 0$. In the proof of Proposition 2, I showed that $\frac{\partial u_S^1}{\partial \sigma} < 0$ and $\frac{\partial u_S^{\theta_2}}{\partial \sigma} > 0$, ensuring this condition is satisfied. Then, because there is a unique $\sigma^*(\beta)$ and unique $\omega(\beta)$ for all $\beta \in (\underline{\beta}, \bar{\beta})$, $\sigma^*(\beta) = \omega(\beta)$. Moreover, the local continuity of $\omega(\beta)$ on a connected space implies global continuity on the space. ■

I now prove Proposition 5.

Proof. Suppose $\theta_2 \in (0, \frac{1}{2})$ $\beta \in (\tilde{\beta}, \bar{\beta})$. Then an IE exists where there is a deterrence benefit from verification (described by Proposition 2(a)) and an IE exists where there is not (described by Proposition 2(b)). In the latter, the expected informational effect of verification is

$$\frac{(2 - \theta_2)^2}{18}. \quad (9)$$

In the former, the expected informational effect of verification is

$$\frac{(2 - \sigma^*)\sigma^* + \theta_2^2(1 - \sigma^{*2})}{3(2 - \sigma^*)(1 + \sigma^*)}. \quad (10)$$

(10) is increasing in σ^* :

$$\frac{\partial(10)}{\partial \sigma^*} = \frac{1}{3} \left(\frac{1}{(1 + \sigma^*)^2} - \frac{\theta_2^2}{(2 - \sigma^*)^2} \right) > 0$$

Moreover, for $\sigma^* = 1$, $(10) = \frac{1}{6}$. Hence, if

$$\begin{aligned} (9) &< \frac{1}{6} \\ \Leftrightarrow \bar{\theta}_2 &\equiv 2 - \sqrt{3} < \theta_2, \end{aligned} \tag{11}$$

then for σ^* sufficiently close to one, $(10) > (9)$. Note, $\bar{\theta}_2 < \frac{1}{2}$.

Fix $\theta_2 > \bar{\theta}_2$. Recall from the proof of Proposition 2 that for all $\beta \in (\underline{\beta}, \bar{\beta})$, there is a unique σ^* that solves (6). Taking the limit of $a^*(\sigma, \theta)$ and $b(\sigma, \theta)$ as $\beta \rightarrow \bar{\beta}$, it is clear that (6) is uniquely solved by $\sigma^* = 1$ for $\beta = \bar{\beta}$. Proposition 2 implies that for all $\beta \in (\tilde{\beta}, \bar{\beta})$, in the IE described by Proposition 2(b) $\sigma^* \in (0, 1)$. Moreover, Lemma A implies $\sigma^*(\beta)$ is continuous. Hence, for β sufficiently close to $\bar{\beta}$, $(10) > (9)$. ■

7 Supplementary Appendix: Additional Results

7.1 Example Equilibrium Without Assumption 1

Suppose $\theta \in \Theta = \{0, \frac{1}{3}, \frac{2}{3}, 1\}$, $h(\theta) = \frac{1}{4}$, $\kappa \sim \mathcal{U}[0, 1]$, and $\beta < \frac{8}{81}$. Consider the following strategy for the sender: the type-0 and type- $\frac{2}{3}$ senders lie with $m = 1$, the type- $\frac{1}{3}$ sender lies with $m = \frac{2}{3}$, and the type-1 sender tells the truth. This violates Assumption 1 since sender's lying strategy is not independent of θ .

The type- $\frac{1}{3}$ sender's expected utility from lying with $m = \frac{2}{3}$ is $\frac{1}{3}$ since the message is only sent by the type- $\frac{1}{3}$ sender and is not verified. The type 0 and type- $\frac{2}{3}$ senders' expected utility from lying with $m = 1$ is also $\frac{1}{3}$ since $\Lambda(1) = \frac{8}{81}$, and so if $\beta < \frac{8}{81}$, $m = 1$ is verified with certainty and $a^*(1, f) = \frac{1}{3}$. Finally, the type-1 sender's expected utility from telling the truth is 1.

It remains to consider whether any type of sender has a profitable deviation. By Assumption 2, $m = 0$ is believed. Thus, no type of sender has a profitable deviation $m = 1$. Moreover, no type of sender who lies has a profitable deviation to lying with a different message in the support of the sender's strategy, and the type-1 sender does not have a profitable deviation to lying. Since $a^*(m, v)$ is not constant, this is an IE.

7.2 Example Equilibrium Without Assumption 2

Suppose $\theta \in \Theta = \{0, \frac{1}{2}, 1\}$, $h(\theta) = \frac{1}{3}$, $\kappa \sim \mathcal{U}[0, 1]$, and $\beta = \frac{1}{4}$. Consider the following strategy for the sender: The type-0 and type- $\frac{1}{2}$ senders lie with $m = 1$ and the type-1 sender tells the truth. Given such a strategy, $m \in \{0, \frac{1}{2}\}$ are off the equilibrium path. Assume that in response to such a message, the receiver believes $\theta = 0$. As such, he chooses $a^*(m, v) = 0$. This violates Assumption 2 since the receiver does not believe messages that are not in the support of the sender's strategy.

The type-0 and type- $\frac{1}{2}$ senders' expected utility from lying is $\frac{3}{8}$, and the type-1 sender's expected utility from telling the truth is $\frac{3}{4}$.

It remains to consider whether any type of sender has a profitable deviation. Since deviating to $m \in \{0, \frac{1}{2}\}$ yields a payoff of 0, the sender never deviates. Moreover, since $a^*(m, v)$ is not constant, this is an IE.

7.3 Existence of an IE

Proposition B. *Define $\check{\beta}$ and $\dot{\beta}$ as in (12) and (13). An IE exists if at least one of the following is satisfied:*

(a.) $\mathbb{E}[\theta] \geq \theta_{N-1}$ and $\beta \geq \check{\beta}$.

(b.) $\beta \leq \dot{\beta}$.

Proof. Consider a strategy for the sender where she lies if $\theta \in \{\theta_1, \dots, \theta_{N-1}\}$ by reporting $m = \theta_N$ and tells the truth if $\theta = \theta_N$. Then when the sender lies, her expected utility is $\pi(\theta_N)\mathbb{E}[\theta|\theta \in \{\theta_1, \theta_{N-1}\}] + (1 - \pi(\theta_N))\mathbb{E}[\theta]$, where

$$\pi(\theta_N) = G\left(\frac{\text{Var}(\theta_N|c=0) - (1 - \phi(m))\text{Var}(\theta_N|v=f)}{\beta}\right),$$

using the fact that G is the CDF of the distribution of κ , and $\pi(m)$ denotes the probability the receiver verifies m . The only effect of β on $\pi(\theta_N)$ is through the denominator, which means $\pi(\theta_N)$ is decreasing in β . Hence, if $\mathbb{E}[\theta] > \theta_{N-1}$, there is a unique β such that

$$\pi(\theta_N)\mathbb{E}[\theta|\theta \in \{\theta_1, \theta_{N-1}\}] + (1 - \pi(\theta_N))\mathbb{E}[\theta] = \theta_{N-1}. \quad (12)$$

Denote this β as $\check{\beta}$. Then for all $\beta \geq \check{\beta}$, $u^{lie} \geq \theta_{N-1}$.

It remains to consider whether any type of sender has a profitable deviation. By Assumption 2, deviating to an off the equilibrium path message yields a payoff weakly below u^{lie} . Moreover, since $m = \theta_N$ is verified with a positive probability, the type- θ_N sender's expected utility from telling the truth is strictly greater than u^{lie} . Hence, there are no profitable deviations.

Now consider a strategy for the sender where she lies if $\theta = \theta_1$ and tells the truth if $\theta > \theta_1$. In particular, when she lies, she uses a strategy where the probability she lies with $m = \theta_i \in \{\theta_2, \dots, \theta_N\}$ is $\sigma_i \in (0, 1)$. A vector of lying probabilities, σ induces a vector of probabilities of verification, π , where, for a particular $\pi(m) \in \pi$, $\pi(m) = G(\frac{Var(\theta|m, c=0)}{\beta})$. Fix σ . The only effect of β on $\pi(m)$ is through the denominator. So there exists a β such that

$$1 = \min(\pi(m) \in \pi). \quad (13)$$

Denote this β by $\dot{\beta}$. Moreover, for all $\beta \leq \dot{\beta}$, the receiver verifies each m in the support of the sender's strategy with certainty.

It remains to consider whether any type of sender has a profitable deviation. By Assumption 2, deviating to $m = \theta_1$ yields a payoff of θ_1 , which is the type- θ_1 sender's expected utility in equilibrium. Moreover, since $m > \theta_1$ are verified with a positive probability, the senders of type $\theta > \theta_1$ have expected utility strictly greater than θ_1 . Hence, there are no profitable deviations. ■